Two-Pass 3D Reconstruction for Line-of-Sight Audits at Crosswalks

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Abstract

We want to replay a real intersection in true 3D over time and ask a simple safety question: could this person at the curb actually see that approaching car. We build a 3D snapshot for each moment from eight synchronized cameras. With that finished world we compute from a pedestrian's eye to the vehicle and measure when and how much it was visible and what blocked it.

1. Introduction

This project creates a complete, time-consistent 3D video of a traffic intersection from eight synchronized cameras to audit the line-of-sight between pedestrians and approaching vehicles [6]. A two-pass method first generates a 3D snapshot for each moment, then temporally fuses partial views of dynamic subjects into complete 3D models so that occluded geometry is recovered before safety analysis, extending multi-view reconstruction ideas popularized by neural radiance fields [6]. The final interactive 3D scene allows for precise ray-casting analysis to determine when a pedestrian's view of any vehicle travelling toward the intersection was obstructed, closing the gap between manual sight-distance checks and data-driven audits [1].

1.1. Problem Statement

At urban intersections, pedestrians at the curb often face blocked views of approaching vehicles due to occlusions like parked cars, poles, or other obstacles. This visibility issue creates a serious safety risk, impairing people's ability to judge traffic conditions before crossing and contributing to preventable accidents documented in intersection safety reviews [5]. In the United States alone, 7,388 pedestrians were killed in traffic crashes in 2021, with limited sight distance called out as a contributing factor in urban corridors [7].

1.2. Research Significance

Solving this matters for safer streets and better urban design. By building time-consistent 3D models from multi-camera data, this project audits line-of-sight (LoS), a core principle of transportation safety [2], between pedestrians and vehicles, pinpointing exact blockers that hide dangers. These insights can drive real fixes, such as removing or redesigning obstacles, adding signs, or implementing smart traffic controls. Broader impacts include fewer pedestrian-vehicle crashes, improved transportation planning, and more accessible cities. Quantitatively, enhanced LoS could reduce accident rates, cut costs in healthcare and insurance, and minimize disruptions in daily mobility, leading to more resilient and people-centered urban environments while keeping documentation compatible with transportation design guidance [1].

2. Proposal

2.1. Related Work

Today, most visibility or safety audits at intersections are done manually or in 2D. City planners, engineers, or researchers usually look at 2D video footage, static camera views, or CAD maps to perform LOS audits [1].
The issue is that real-world visibility is 3D and dynamic: people and vehicles move, change orientation, and are partially hidden by other moving or static objects. Traditional methods cannot capture these occlusions correctly, especially when multiple cameras or viewpoints are involved, so planners often miss hidden hazards.

2.2. Proposed Method

We will build a 3D Line-of-Sight (LoS) Visibility Audit system that reconstructs intersections in full 3D and tests whether pedestrians at the curb can actually see approaching vehicles (and vice versa).
Use multi-view video data (8 synchronized cameras) to reconstruct a metric 3D model of the scene using Fast-VGGT [10] for per-timestamp depth and pose estimation. Apply a two-pass reconstruction method: the first pass builds per-timestamp 3D scenes and the second pass fuses later frames of each moving subject to create complete 3D models that fill in missing sides. To link the same subjects across time we use YOLO [9] (object detection) and

SAM [4] (segmentation and tracking) for object detection and segmentation.

Perform line-of-sight testing by casting rays from a pedestrian's eye point toward intersecting vehicles in the 3D scene, checking for occlusions from static and dynamic objects. This will allow us to produce a LOS audit, showing exactly where and when visibility is lost and which design elements or parked vehicles cause the issue.

The system outputs interactive reports that match the camera views used in safety studies while attaching 3D overlays that pinpoint root causes of occlusion.

2.3. Why it works?

The system succeeds because the two-pass reconstruction gives us more complete and realistic 3D geometry than traditional single-pass or 2D methods. By fusing views and frames, we recover hidden sides of vehicles and pedestrians, which is essential for accurate visibility testing. Proven tools like Fast-VGGT [10] (for 3D reconstruction), YOLO + SAM [4, 9] (for detection/segmentation), and ICP [3] (for temporal alignment) give reliable and reproducible results.

In short, the method directly addresses the lack of full 3D, time-consistent visibility in current audits and leverages modern computer vision models to create a realistic, measurable, and interpretable audit system.

3. Methodology

We begin by cleaning the footage, skipping black sync frames and choosing a short well lit window (15–30 seconds per scenario). We anchor scale with a simple scene prior such as crosswalk stripe spacing so distances are meaningful and consistent with roadway design drawings.

Pass one asks what we saw right now. The eight images at a time go to Fast-VGGT [10], which returns depth for each view and camera poses with reported $<$ 5 cm RMSE on multi-view datasets. The slice mixes the road and the movers, so we separate them. YOLO [9] marks people, bikes, and vehicles at real-time rates (45 FPS on 640 px frames), and SAM [4] turns each box into a crisp mask. We split the slice into a static background and per-object point clouds. A simple 3D tracker assigns a stable identity to each subject using motion, proximity, and a compact appearance cue.

Pass two lets the future complete the past. For each tracked vehicle we recenter the masked points into the vehicle frame using its pose timeline. Short, robust alignment stitches the pieces into a single canonical car, leveraging ICP-style fitting [3] with a 2 cm convergence tolerance. Pedestrians are represented with a simpler non-rigid proxy, which is enough for visibility. We then replay the clip and place the completed car into the early frames where only a bumper had been seen. Frame by frame the world becomes a consistent 3D video.

The line-of-sight audit is then a direct query. For a pedestrian we set an eye point around fixed heights' above the road to address diversity. We sample key points on each approaching vehicle and cast rays through the scene at 1 cm step size. If a ray hits something first the target is not visible. From these rays we read three values that matter for safety: when the vehicle first became visible, how much of it was visible over time, and how long it stayed hidden while approaching.

Feasibility is practical. A focused prototype fits in three weeks on a single cloud GPU. Week one delivers per-timestamp reconstruction, instance masks, tracking, and a viewer of the 3D slices. Week two delivers vehicle fusion across time, placement back into earlier frames, and week three integrates the LoS audit with initial quantitative measurements.

4. Planned Experiments

4.1. Data

Dataset: StreetAware: A High-Resolution Synchronized Multimodal Urban Scene Dataset [8]

We use sample sequences captured by four corner sensor pods (8 synchronized RGB cameras).

4.2. Evaluation Metrics

- **End-to-End Line-of-Sight (LoS) Accuracy**
  Visibility score (0-1): fraction of the vehicle's projected surface whose depth lies in front of the scene depth within $\pm$30 cm. Clear $\geq$ 0.6 Partial 0.1-0.6 Complete $\leq$ 0.1
  Occluder Attribution: Top-1 accuracy for identifying the blocking object (parked car, pole/sign, moving vehicle, tree/foliage).
- **3D Completeness and Quality Metrics**
  The 3D completeness and quality metrics assess the reliability of reconstructed geometry for LOS analysis. Voxel Occupancy Coverage (VOC) measures the percentage of occupied voxels within a 5 cm grid, indicating reconstruction density; and Multi-View Reprojection Consistency (MVRC) quantifies depth consistency across multiple views, also serving as a confidence filter [12].

4.3. Comparative Baselines

Backbones: FastVGGT (primary) vs PI3 (baseline for quality/speed trade-off) [10, 11]

Confidence Usage: With vs without MVRC-based masking to demonstrate robustness against noisy geometry.

2D Control (No 3D): Simple image-plane line test between pedestrian and vehicle bounding boxes, labeling visible when no other box lies between, included to illustrate the necessity of 3D reasoning.

4.4. Hypotheses and Experiments

Hypothesis: A Two-Pass, confidence-filtered 3D reconstruction (actor completion across time + MVRC masking) produces more accurate LoS decisions and occluder identification than single-pass or 2D heuristics on StreetAware multi-view scenes [8].

Experiments:

- **E1**: Establishes a baseline by fusing multi-view FastVGGT/PI3 outputs into a metric-scaled world point cloud and evaluating runtime and MVRC [10–12].
- **E2**: Performs two-pass actor completion using tracking, segmentation, and ICP-based fusion to enhance VOC and MVRC [3, 4, 9, 12].
- **E3**: Builds an integrated 3D scene video combining static background and dynamic actors for qualitative inspection.
- **E4**: Conducts the main LoS audit by casting rays from pedestrians to vehicles and evaluating visibility scores, benchmarking against the 2D control.
- **E5**: Tests robustness under small pose perturbations, comparing FastVGGT vs PI3, single-frame vs two-pass, and 3D vs 2D control approaches [10, 11].

References

[1] American Association of State Highway and Transportation Officials. *A Policy on Geometric Design of Highways and Streets*. AASHTO, 7th edition, 2018. 1

[2] American Association of State Highway and Transportation Officials. *A Policy on Geometric Design of Highways and Streets*. AASHTO, Washington, D.C., 7th edition, 2018. Commonly referred to as the AASHTO Green Book. 1

[3] Paul J. Besl and Neil D. McKay. A method for registration of 3-d shapes. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 14(2):239–256, 1992. 2, 3

[4] Alexander Kirillov, Eric Mintun, Nikhila Ravi, Hanzi Mao, Chloe Rolland, Achleshwar Gustafson, Tete Xiao, Spencer Whitehead, Alexander C. Berg, Wan-Yen Lo, Piotr Dollár, and Ross Girshick. Segment anything. *arXiv preprint arXiv:2304.02643*, 2023. 2, 3

[5] Dominique Lord and James A. Bonneson. Safety impacts of signalized intersection design features: A literature review. *Transportation Research Record*, 2023(1):43–52, 2007. 1

[6] Ben Mildenhall, Pratul P. Srinivasan, Matthew Tancik, Jonathan T. Barron, Ravi Ramamoorthi, and Ren Ng. Nerf: Representing scenes as neural radiance fields for view synthesis. In *Proceedings of the European Conference on Computer Vision (ECCV)*, pages 405–421, 2020. 1

[7] National Highway Traffic Safety Administration. Traffic safety facts: Pedestrians 2021 data, 2023. DOT HS 813 450. 1

[8] Yurii Piadyk, Joao Rulff, Ethan Brewer, Maryam Hosseini, Kaan Ozbay, Murugan Sankar, and Srimat Chakradharand. StreetAware: A High-Resolution Synchronized Multimodal Urban Scene Dataset, 2023. 2, 3

[9] Joseph Redmon, Santosh Divvala, Ross Girshick, and Ali Farhadi. You only look once: Unified, real-time object detection. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 779–788, 2016. 1, 2, 3

[10] You Shen, Zhipeng Zhang, Yansong Qu, and Liujian Cao. Fastvggt: Training-free acceleration of visual geometry transformer, 2025. arXiv preprint arXiv:2509.02560v1, 02 Sep 2025. 1, 2, 3

[11] Yifan Wang, Jianjun Zhou, Haoyi Zhu, Wenzheng Chang, Yang Zhou, Zizun Li, Junyi Chen, Jiangmiao Pang, Chunhua Shen, and Tong He. π^3: Scalable permutation-equivariant visual geometry learning, 2025. arXiv preprint arXiv:2507.13347v1. 2, 3

[12] Yao Yao, Zixin Luo, Shiwei Li, Tian Fang, and Long Quan. Mvsnet: Depth inference for unstructured multi-view stereo. *Proceedings of the European Conference on Computer Vision (ECCV)*, pages 767–783, 2018. 2, 3